



Module 17

Mental Training

Whether it relates to the perception of pain, a governor that heavily influences pacing, or the even the desire to run, the brain has a large influence on how the body regulates itself, which in turn relates to human performance.

Understanding on a small scale how the mind/body connection functions will enable you to be a better and more effective coach.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Mind-body connection
- Sport-specific uses of mental training
- Confidence
- Performing under pressure
- Mental imagery
 - Internal
 - External
- Mental fatigue
- Desensitization
- Placebo effect
- Pain threshold versus pain tolerance
- Mental burnout
- Concentration/Focus
- Motivation

❖ TERMINOLOGY

Mental Imagery: The process by which one imagines or visualizes performing an activity. This creates the same neural patterns in the brain as occurs during the actual physical activity.

Mental Burnout: Refers to a phase in which an athlete has no desire to train or compete over a long period.

Confidence: A feeling or consciousness of one's powers or of reliance on one's circumstances (453).

Desensitization: Strategy used to reduce and/or eliminate phobias through repeated and increased exposure to the phobia.

❖ MENTAL TRAINING ELEMENTS

The mental side of training is often validated as being extremely important, but when it comes time to set up a training program, it is typically left out or put at the very bottom of the list.

- How many times do you hear of child prodigies who burn out and are never heard from again?
- How often do you hear the phrase “mentally tough” in regard to athletes?
- Why do most top athletes distinguish themselves by being able to perform under extreme pressure versus cracking like “regular” athletes?
- Most athletes will likely agree that confidence is a major factor in performing well.

While it is true that some athletes are born with exceptionally high degrees of mental “toughness,” the mental side of sport can be trained and developed like the physical side.

Not a lot of professional, peer-reviewed studies relate to the mental aspect of distance running, specifically as it pertains to mental strategies.

Mental training is commonly used for:

- Familiarization with an environment (especially new environments, such as race venues)
- Motivation
- Skill development and perfection
- Reduction of negative thoughts
- Establish or reestablish focus
- Visualize success
- Reduction of anxiety

The Mind-Body Connection

A strong correlation exists between the body and the mind. Two examples of this are pain and panic attacks:

Pain

When an individual touches a hot object with a hand, nerve receptors called *nociceptors* send information to the brain that is interpreted as pain. While the origin of pain may be in the hand, the brain interprets and registers this nerve signal as pain.

Panic Attacks

A panic attack is a sudden and intense fear that manifests itself with symptoms very similar to that of a heart attack, but without a real cause of danger being present. The origin of the anxiety is the brain, and it is manifested as physical symptoms.

The purpose of these examples is to demonstrate the strong connection between the mind and body.

CONFIDENCE

Confidence is perhaps one of the biggest factors of success. Confidence can be defined as (453):

A feeling or consciousness of one's powers or of reliance on one's circumstances.

While confidence can be ascertained in many ways, it is also a trait that can be situation specific. For example, a successful lawyer might feel right at home and 100 percent confident in a courtroom but might be intimidated at the start line of a running race. It is the responsibility of a coach to help clients feel confident. However, this is possible only if coaches are confident in their abilities and expertise.

Success and confidence are cyclical in nature, and one builds on the other. In order to have success, people need to be confident in their abilities. When success is achieved, a client's confidence is boosted even higher, and the individual is more likely to have continued success in the future. While not all outcomes of training and racing are going to be positive, having a coach instill confidence in clients will enable them to stay on the right mental path to achieve success over the long term.

Following are some techniques that can help to build confidence in your clients:

1. Remind your clients that they have prepared properly and to stick to the game plan in regard to pacing, nutrition, etc.
2. Because they prepared properly for the event, they should have faith in their preparation and fitness level.
3. If clients are anxious, remind them of other areas in which they perform well under pressure so they know that this is situation specific and does not reflect on them as a whole.
4. Remind them that they can handle whatever situation they encounter during a race.
5. If a client is really nervous, have the person calm down by taking long, deep, slow breaths to help regulate respiration and heart rate. Assuming your client is proficient at mental imagery, have the individual imagine performing at a high level. In many instances, clients experience prerace anxiety. Once the event begins, most of the time the anxiety levels decrease.

6. Positive thinking. Clients should not dwell on negative feelings, as this will have a toxic effect on performance and the overall enjoyment of the event. Positive self-talk helps to facilitate positive thinking.

Are You Expecting to Fail?

Most seasoned endurance athletes have at one time or another “hit the wall.”

While glycogen depletion is the physical cause, two studies found that one’s expectation to hit the wall may be associated with doing so during marathons (703, 704).

PERFORMING UNDER PRESSURE

Performing under pressure or stress has a lot to do with confidence in one’s self. One trait that sets successful athletes apart from others is their ability to thrive under pressure. Keep in mind that pressure is subjective. For a new runner, pressure might be showing up to the start of a race, while for a professional runner, pressure might be going head-to-head in the final kilometer of a race for the win.

How one deals with and interprets pressure has a lot to do with physical performance. While you are most likely not a psychologist or psychiatrist, this does not mean that you cannot affect to some degree how your clients think and view themselves.

An athlete might be extremely successful in practice and training sessions but not in competition. This is often termed *cracking under pressure*. This has little to do with a client’s physical level of conditioning or preparedness, but rather a client’s inability to perform when there is something at stake. During practice, win, lose, or draw, there are no consequences for a poor performance – or at least not as much as with an actual competition. Because they feel this pressure to perform, athletes often become hesitant during a race and overthink their strategy. This is often because they worry about pushing too hard and blowing up. As a result, they may be too conservative and perform poorly.

The opposite situation exists where an individual goes all out, blows up, and has a terrible race. Neither of these race strategies is correct and both can be remedied by taking the time to prepare an athlete mentally for competition.

Mental training is an incredibly broad and vague term that can be broken down much further. For the purposes of this certification, the following areas of mental training will be examined:

1. **Mental Imagery (internal/external)**
2. **Burnout**
3. **Motivation**
4. **Mental Fatigue**
5. **Pain Tolerance vs. Pain Threshold**

❖ MENTAL IMAGERY

The theory behind mental imagery is that when individuals imagine or visualize performing an activity, they create the same neural patterns in their brain as occurs during that physical activity. In other words, the goal of mental imagery is to train the brain to teach the muscles, from a neuromuscular standpoint, how to perform and function.

The regions of the brain responsible for motor control are the same regions responsible for conscious thought and control over these motor regions. Therefore, the areas of the brain that create physical movement are the same areas that are activated during mental imagery, although without physical movement (151).

Functional magnetic resonance imaging (fMRI) has shown that when performing mental imagery, the amplitude of brain waves used for motor skills and electrical stimulation of muscles are activated (116).

Neural pathways are tracts that connect one part of the nervous system to another part. Through mental imagery, neural pathways can be created, and the more a pathway is used, the stronger it becomes (148).

There are two primary categories of mental imagery: *internal* and *external*.

Research

The first documented research on mental imagery in regard to how it is performed and its relative effectiveness was in 1894 by W.B. Carpenter (263). This theory was called the *psychoneuromuscular theory*. While many theories have been introduced to help determine the most effective way to perform mental imagery, there has not been a singular theory that completely explains the effectiveness of mental imagery through a particular rehearsal mechanism (264). Therefore, it could be theorized that any number of mental imagery strategies could be effective.

A 1992 study by Anne Isaac examined the influence of mental imagery on sports skills, specifically, the trampoline (149). In this study, Isaac assessed 78 subjects who were classified as novice and expert trampolinists. These groups were broken down further by those who had poor and good imagery skills. Two primary findings came out of this study:

1. Within groups of the same trampoline skill level, those who had higher imagery skills showed more improvement than those who had lower imagery skills.
2. The group that performed mental imagery showed significant improvement over the group that performed mental tasks (e.g., math problems).

A 1998 study by Roure et al. examined the autonomic nervous system response to mental imagery and, more specifically, if sports performance was improved by imagery. The study assessed an individual's ability to pass an opponent's volleyball serve to a predetermined teammate – both physical and mentally. The assessment revealed a strong correlation between the autonomic nervous system (ANS) response and physical and mental tasks. The group that performed mental imagery in addition to the physical task performed better than the “physical task only” group (150). Based on this information, Roure theorized that mental imagery could help develop organized thoughts or behaviors without conscious thought during physical training.

Another study by Australian psychologist Alan Richardson found that mental imagery was also beneficial in improving athletic performance. This study looked at three different groups of students. The first group practiced basketball free throws for 20 minutes per day for 20 days. The second and third group shot free throws only on the first and last day of the 20-day period. However, the second

group spent 20 minutes per day visualizing shooting free throws. The result was that the first group (physical free-throw shooters) improved 24 percent and the second group (mental imagery) improved 23 percent – almost the same amount (762)!

When reviewing the results from studies that seek to establish a correlation between mental imagery and athletic performance, there seems to be one constant: mental imagery does not enhance physical performance as much as physical training. However, mental imagery in conjunction with physical training elicits better results than physical training alone.

Quality over Quantity

It is not enough just to *imagine* an activity or movement. Like physical training, the quality of the imaging is directly related to its effectiveness. Low attention to detail while performing mental imagery will result in a reduction of the effectiveness of imagery on motor performance.

It is better to have fewer mental imagery sessions that are executed with complete concentration and focus than to have more sessions with low levels of concentration and focus.

The level of concentration and focus required for a successful mental imagery session is substantial. It takes a lot of training and practice to become proficient at mental imagery. There are often areas in which a client is unable to visualize, while other areas might come easily. Like physical training, a focus on the weaker areas is advisable.

The goal is to be able to visualize performing a task with 100 percent accuracy and, thus, have control over the imaging.

INTERNAL IMAGERY

Internal mental imagery (*internal association*) is often categorized into two areas:

1. How one feels when doing a particular activity (physical sensations)
 - a. e.g., imagining how tremendous exertion is going to feel so that the discomfort does not come as a shock.

- b. helps to prepare for the sensations of an event such as temperature, fatigue, effort level, etc.
- 2. Visualize performing an activity from an athlete's own perspective (e.g., runner visualizing running at race pace with lots of surrounding runners).

EXTERNAL IMAGERY

External imagery occurs when runners see themselves from the perspective of someone else or from a video camera (third person). This is extremely helpful when discussing the proper form of running. By imaging what one knows to be correct form, a runner will be better able to attain and correct his or her form if it breaks down. While this is a valuable tool at any stage in the training process, it is especially useful when learning a new skill.

It is imperative to be able to visualize an activity correctly. If clients are unable to see themselves performing an activity or movement correctly, they should continue working on their mental imagery until they can mentally perform the activity with correct form. Once this is accomplished, it is advised to continue mental imagery sessions to reinforce the correct mental image of the activity.

HOW, WHEN, WHERE?

How?

Research by Currie and Ravenscroft (1997) found that for mental imagery to be most effective, imagining (i.e., imaging) the skill and physically performing the skill must occur close together from a time perspective (152). An example of this would be performing five physical attempts, followed by five mental attempts, and finishing with five more physical attempts.

The goal of mental imagery is to make the mental picture as detailed as possible. This means that in addition to visual stimuli, other senses such as hearing, smell, taste, and overall body awareness are often integrated into imaging sessions. The more “real” a mental imagery session is, the more productive it will be.

Learning how to perform mental imagery is like learning how to swing a golf club properly. If one were to try to identify and correct all the areas of a golf swing at

the same time, failure is all but guaranteed. Because of the many components that can be identified during a mental imagery session (i.e., form, breathing, body awareness, hearing, etc.), it is advised to break them up and focus on each individually. Only when these individual areas are perfected should all the areas be integrated.

Association

Associating words that denote feeling or form often aids in establishing correct images and therefore correct physical actions. An example of an association would be thinking of “floating” to help with one’s running gait toward the end of a race.

Visual Cues

Using video of a client or another athlete performing a specific task with correct form can assist in reinforcing what the correct form is supposed to look like.

Garbage in, garbage out is a computing phrase that means if you put bad data into a program, the program will give bad results. This is very applicable to mental imagery. If your client’s idea of correct form is actually poor form, as a result of the flawed mental imagery, the individual’s physical form will very likely be incorrect.

Video use does not have to be limited to assessing only form. For example, if your client is extremely anxious about the start a running race, you could source video of a race start. This will allow your client to appreciate what the start will be like and to prepare for it mentally.

When?

If mental imagery is being done for the purpose of a specific event, mental imagery training must take place well before it. Additionally, mental imagery must be practiced regularly for it to be truly effective – “use it or lose it.”

Those skilled in mental imagery will likely be able to perform imagery sessions at various times (e.g., in training, right before an event, during the event).

Mental imagery also has value during an event, specifically for improving form. Breakdown of form is most typically associated with fatigue. Through the use of external mental imagery, an individual can visualize how the form is supposed to

look and adjust accordingly based on current form/postural deviations. Following are common postural/form issues that arise when fatigued:

- Scapular elevation
- Lumbar and cervical spine extension
- Run with less gait control – typically the stride length lengthens in front of a runner

Through the use of mental imagery, these areas can be identified and ideally corrected.

Where?

Initially, mental imagery should be practiced in an area with minimal external stimuli. As one becomes more proficient, environments with external stimuli can and should be introduced.

DESENSITIZATION

The sport of running has some elements that can cause fear, especially for those new to the sport. Some of the more common areas that runners fear are:

- Tripping and falling
- Race start
- Pacing
- Hitting the wall (e.g., bonking)
- Failure

It is the responsibility of a coach to work with clients to acknowledge and reduce their anxiety and fears regarding running.

Desensitization is a technique that was developed by Mary Cover Jones, a psychologist, considered the *mother of behavior therapy* (348). The purpose of desensitization is to reduce and/or eliminate phobias through repeated and increased exposure to a phobia. Mental imagery can assist in the desensitization process.

Through mental imagery, a client can envision a phobia within a controlled environment. Let us take the start of a running race as an example:

Using internal imagery, a client envisions himself or herself standing among many other runners awaiting the start. A client might initially just focus on this mental picture until not feeling any anxiety. As a client progresses through the desensitization process, the individual would imagine hearing the gun go off, starting to run, and getting up to the desired pace. The more vivid and realistic the imagery (e.g., sounds, sights, smells), the more beneficial it will be.

You are encouraged to read more on the topic of desensitization, especially if you have a client or clients who have phobias regarding one or more aspects of running. If you do not feel capable of or are uncomfortable working with a client regarding desensitization, you are advised to refer the individual to a psychologist or a phobia specialist who can assist.

MENTAL IMAGERY SUMMARY

- Mental imagery will benefit a runner if done properly and with frequency.
- Mental training does not replace physical training.
- By itself, mental imagery is beneficial. When done in conjunction with associated physical activity, it is more beneficial.
- As most runners do not perform mental imagery, this should be viewed as a “secret weapon” in your client’s arsenal of training methodologies.
- Mental imagery can be performed during both training and racing.
- Desensitization is a way to decrease or eliminate a phobia through repeated exposure to a phobia. Mental imagery can help in this process.

❖ MENTAL FATIGUE

Mental fatigue can negatively influence physical performance. A 2009 study by Marcora et al. examined this relationship (661). Participants in this study performed strenuous physical activity when mentally fatigued and mentally rested. Below are the results:

- Participants stopped exercising 15 percent sooner when mentally fatigued.
- While both participant groups (mentally fatigued and nonfatigued) stopped at the same perceived exertion level (RPE), the fatigued group started at a higher RPE level than the nonfatigued group.
- Cardiorespiratory and muscle measurements did not vary between the two groups.

Marcora et al. theorized that potential reasons for the results were:

- Mental fatigue lowers the brain's inhibition against quitting.
- Mental fatigue affects dopamine levels of the brain, which is a chemical that affects an individual's motivation and effort.

Potential Implications for Runners

Mental fatigue, regardless of the cause, may increase the perceived effort of a runner and thus reduce physical performance (661).

While not all potential causes of mental fatigue are avoidable (i.e., travel to running race), by reducing the number of potential causes of fatigue before and during a running event, the better the outcome will likely be. Following are examples of areas to address with your client.

- If traveling to an event, make sure all arrangements (i.e., hotel/flight/car/restaurant reservations) are confirmed prior to traveling.
- Know the course profile so that nothing comes as a surprise.
- Train in all weather conditions to be mentally prepared for any scenario.
- Get plenty of sleep.
- Have a pacing plan for the race.
- Consume only familiar foods/drinks before and during the race – nothing new on race day!
- Get to the race registration and start line in plenty of time.
- Implement a proper training plan to ensure confidence on race day.

❖ PAIN THRESHOLD vs. PAIN TOLERANCE

What makes a great athlete a great athlete? Is it better physiology, better body mechanics, or a better ability to deal with pain? Probably all of the preceding. Following are results from research performed in the area of pain threshold and pain tolerance (345, 346, 347).

- Athletes and nonathletes have similar *pain threshold levels* (the point at which pain is felt)
- Athletes are able to tolerate pain better than nonathletes

What the research did not conclusively determine is why athletes have higher pain tolerances. Following are possible reasons for higher pain tolerances in athletes versus nonathletes (347).

1. Athletes are more motivated to perform at higher levels of intensity than nonathletes
2. Resistance to pain can possibly be learned over time
3. Better coping skills to deal with pain (520)
4. Genetics

It could also be deduced that the higher one's tolerance is to pain, the greater the chance for injury. While some pain (good pain) is the result of high-intensity workouts, other pain (bad pain) is the result of musculoskeletal injury. It could be assumed that an individual who has a high pain tolerance could train or race through the warning signs of potential injury (i.e., pain). This could increase the chance for injury as well as the severity of an injury compared with someone who reduces, modifies, or stops training with the onset of musculoskeletal pain.

A 2013 study by Geva et al. found that not only do athletes (triathletes were assessed in this study) have a better tolerance to pain than nonathletes, they have a reduced fear of high levels of pain (734). This likely contributes to athletes' ability to pace themselves for long periods of time at high intensity levels. This relates to the study by Marcora et al. that found that mental fatigue limits one's tolerance to exercise (735). In other words, pain can increase mental fatigue, which in turn likely results in reduced pain tolerance.

This section directly relates to the *perceived intensity assessment* (PIA) in the Client Intake section of the *Training Program* module.

❖ BURNOUT

One aspect of mental training that is rarely taken into consideration when coaching athletes is that of mental burnout. All athletes experience mental highs and lows when training. However, when this certification uses the term *burnout*, it is referring to a phase in which an athlete has a reduced or a complete lack of desire to train or compete.

Regardless of the textbook definition, when coaching runners, it is of crucial importance that you are aware of what burnout is and that it can occur to your clients. The goal is to motivate and progress your clients, but without pushing them to the point of burnout. During the coaching process, you will learn how long and hard you can push them before negatively affecting their performance both physically and mentally.

Mental burnout is not just a function of the intensity and/or duration of a training program. Burnout can also result from other things going on in a client's life. For example, if a client is going through a divorce, this stress might mean that he or she cannot tolerate high training workloads because so much energy is channeled toward dealing with the personal strife. Conversely, some clients in this scenario might have more energy as they are using the training as an outlet for their stress. As noted above, the longer you work with clients, the more you will get to know them in regard to what they can handle mentally and physically.

Another common reason for burnout is lack of progress. Whether your client is trying to lose weight or train for a marathon, motivation typically declines when a program is being followed but little to no forward progress is realized. This lack of forward progress is sometimes imagined on the part of a client. This usually occurs because the individual might be looking at only one aspect of the training program. It is the responsibility of a coach to assess the training program as a whole and demonstrate to the client where progress has been made (assuming it has). As running has many quantitative measurements, scheduling performance testing at set intervals in the training process will give an athlete quantitative data to demonstrate progress.

The last concept to understand relative to burnout is forced training sessions. You will have many days when clients will not want to train because they feel tired, unmotivated, lazy, etc.

In many cases, it is advantageous to push through and train because, more often than not, a client will feel refreshed and motivated after a workout. However, when a lack of motivation lasts several days in a row, you need to recognize it as a possible early sign of burnout. Forcing training sessions onto a client during this time will usually have a negative effect, both physically and mentally. It is advisable in these cases to have a client take some time off or do other forms of exercise unrelated to running that lack structure. This provides a mental break needed to come back to training with renewed energy and motivation.

❖ CONCENTRATION/FOCUS

Maintaining concentration during training and racing is important. A lack of concentration can lead to injury, decreased performance, or even a DNF.

A popular way to stay focused is to concentrate on pace. Whether it be heart rate, running pace, or perceived exertion levels, concentrating on pace is a way to stay focused during training and racing. Another popular way to pace and stay focused is to concentrate on one's breathing rate. Regardless of the pacing tool, being able to identify and utilize something to assist with concentration during training and racing is important for most athletes.

A lack of focus can also be due to low blood sugar, especially when exercising for long periods. This is where proper fueling comes into play.

❖ MOTIVATION

It is important to gauge not just the physical aspect of training, but the mental aspect as well. One way to gauge this is to have athletes rate their training motivation level each day. Some athletes can train for months on end, at high intensity levels, without any dip in motivation. However, others may be able to train for only three to four weeks before realizing a decrease in motivation.

What provides an athlete with motivation is often a very individual thing. For some athletes it could be knowing that at the end of a race their family is waiting for them while others could be motivated by setting a *personal record* (PR). The same holds true during the training process in that the sources of motivation are largely individually based. It is important that an athlete has a source of motivation throughout the training process.

Common ways endurance athletes motivate themselves are:

- Break a race or training distance into sections
 - e.g., focus on getting to the end of a particular section, not the end of the race
- Remember the reason why they are training so hard
 - e.g., raising money for a charity
- Want the sense of accomplishment and achievement that comes with finishing a workout or race
- Setting a PR
- Train with a partner, especially for long runs

TRAINING LOGS

Implementation of daily training logs is an important part of the training process not only to chart progress, but also to document trends (both positive and negative). In regard to burnout, clients should note on a scale of 1–10 (10 being the best) how motivated they feel on a daily basis.

Perhaps you begin to see a trend (ex: unmotivated) taking shape every seven weeks or after a bad race. Be aware of trends so that you can work with an athlete and adjust the program accordingly (if suitable). Daily training logs can also identify when an athlete is not recovered. For example, if you notice that a client always rates the session quality as low when the max heart rate is low, it is a sign that the person has not recovered and probably should have taken the day off. Instances like this are why training logs are so important – to spot trends over time and make the necessary adjustments.

PROGRAM FLEXIBILITY

If an athlete is suffering from substantial mental burnout, the program requires adjustment.

Severe mental burnout occurs when an athlete is at such a low point it would be counterproductive to continue training, as this would lead to a further reduction in motivation. All athletes have a point at which they lose motivation; it is a matter of finding out where this point is for a particular client and what the triggers and signs are.

MUSIC

Athletes often use music to help motivate themselves during workouts. Safety issues aside, can music increase the performance of an athlete? Based on research presented here, the answer is yes and no.

Costas Karageorghis, Ph.D., is a sports psychologist and author of the book *Inside Sports Psychology* (505). In this book, Dr. Karageorghis notes that listening to music can improve running performance by up to 15 percent. This is largely due to music's ability to reduce one's perception of intensity/exertion. More specifically, studies by Dr. Karageorghis found that listening to music while running can reduce one's perception of exertion by 10 percent (501, 503). The reasoning behind this is called *dissociation*. Dissociation means that positive mood aspects such as happiness are elevated while negative aspects (e.g., depression) are reduced (504). It must be noted that the positive performance effect of music is typically limited to low to moderate intensities. Once the exercise intensity increases to a high level, the positive effect of music is largely negated (504).

These findings correlate with a study that examined the effect of various music tempos with cycling performance (500). In this study, 12 subjects listened to music that was played at slow, medium, and fast tempos. Subjects pedaled faster when listening to fast-tempo music.

Listening to music while exercising does not have the same effect on all individuals. Athletes who have good self-awareness (i.e., heart rate, muscle tension, posture, breathing, etc.) are not likely to see the same benefit from music as someone who is looking for a distraction to help "get through" a workout (504).

Since music can “artificially” increase one’s intensity past what the individual should be performing at, it can be theorized that listening to fast-paced music could increase the chance for injury and/or overtraining.

As some events do not allow the use of headphones, your clients should not become dependent on music during training.

If running outdoors, it is advised to listen only to earphones that allow ambient noise to be heard over the music, or if using earbud-type headphones, use only one earpiece. If cycling for cross-training, earphones should never be used.

❖ **PLACEBO EFFECT**

Expectations of athletes can have a significant impact on performance, both in training and racing. The most well-researched area in regard to this is the placebo effect.

Most often associated with the medical field, the placebo effect is defined by an individual realizing a perceived or actual improvement, where the stimulus or catalyst is not something that should physically elicit an improvement (754). From the standpoint of increased performance, this essentially means that if an individual believes that a particular training/recovery methodology or product will help, there is an increased likelihood that it will, regardless if the methodology is responsible for increasing performance.

In regard to training and racing, the placebo effect is an important variable to discuss, given its potential effect on training methodologies and equipment.

A recovery technique widely used by athletes and nonathletes alike is Epsom salt baths. Advocates of Epsom salt baths will likely use words and phrases such as “detox,” “cleanse,” and “reduce inflammation and soreness” to describe the benefits. The theory behind Epsom salt baths is that the dissolved magnesium sulfate in the water permeates the skin barrier to help reduce inflammation and pain, thus accelerating the rate of recovery. The problem? There is no research

to support this. Despite this, countless people swear by the positive effects of Epsom salt baths (755).

Countless training methodologies and products claim to increase performance with either limited or no evidence-based research to back up the claims (756). While the genesis of this certification is to focus on the science behind training methodologies and practices, it is important to know and understand that the placebo effect is real. Therefore, assuming that a product or training practice is not potentially injurious or contraindicated in any way for a client, it likely should not be discouraged as it may help your client – or at least not hurt.

❖ STRATEGIES

If you talk to finishers of a marathon, you will likely find that they utilized mental strategies to get through the race. These mental strategies are largely used to combat fatigue for the purpose of finishing the race and, for some, finishing within a specific goal time.

ASSOCIATIVE AND DISSOCIATIVE

In a study of 66 finishers of the 1996 London Marathon, four mental strategies were most prevalent among nonelite runners (662):

Internal Association: Also termed *internal mental imagery*, this focuses on how an individual feels when running.

- Ex: My breathing is labored by I can still get a full breath.

Internal Dissociation: This strategy is essentially a way to distract runners from paying attention to how they're feeling.

- Ex: Imagining themselves on a beach, playing a song in their head.

External Association: Focus outwardly on aspects of the event that relate to one's running performance.

- Ex: Focusing on split times at mile markers, catching a runner in front of them.

External Dissociation: Focus outwardly on aspects of the event that do not relate to one's running performance.

- Ex: Look for friends in the crowd, take cell phone pictures while running.

Of these strategies, *internal association* was the most prevalent strategy while *internal dissociation* was the least prevalent among the runners.

The strategy most associated with “hitting the wall” or “bonking” was *internal dissociative*. This suggests that by not paying attention to how the body is feeling, sensory feedback is blocked or greatly reduced. While on a much lesser scale, this is akin to someone who does not have feeling in his or her extremities putting a hand on a hot object. Since the individual does not have sensory feedback of the heat, the person can easily get burned without knowing it. Therefore, this mental strategy is not recommended.

Interestingly, *internal association* was also linked to higher rates of “hitting the wall.” This suggests that perhaps focusing too much on physical symptoms may exaggerate a runner’s discomfort.

External dissociation was correlated to a later onset of “hitting the wall.” This is likely due to a runner being distracted by nonperformance race factors but still focusing on the race.

Regardless of the mental strategy used by a runner, the primary thing to consider is that an individual’s thought process can and likely will influence running performance and race result.

OTHER STRATEGIES

While the mental strategies one can use are near limitless, Jeff Brown, a psychologist for the Boston Marathon and the US Olympic Committee, outlines several strategies (663):

1. Trust your training

You’ve put in the work, so trust that you’re ready for the event and have fun!

2. Practice race-day routine

Prior to a race, practice your precompetition routine (i.e., waking up, eating certain race-day food, have clothes/shoes ready and laid out). This will familiarize you with the morning race-day process.

3. Eliminate negative thoughts

Think only positive thoughts. Do your best to eliminate the *what-ifs* before a race and focus before and during a race on positive things.

4. Get rid of distractions before and during a race

There are always things that can distract you (i.e., that big presentation when you get back in the office on Monday). However, this is your day that you've been training for. Unless it's a life-or-death distraction, it can wait until you're done with your race. You've trained hard for this event, and it deserves your full attention.

5. Is normal and OK if unexpected things come up

There is no such thing as a perfect race. Whether it's a function of getting slowed behind a runner or having to take a break because of leg cramps, unexpected things *will* happen. Do not let these things derail you. Many of these things you cannot control, so it is best to just acknowledge them, accept them, and move on. Perfection is not required to finish a race or even hit your PR.

6. Talk to yourself

No, this doesn't qualify you as crazy! Self-talk, whether it's audible or not, can help to enhance your performance. Thinking or saying positive words and phrases (e.g., "strong," "light," "float," "I'm strong") can help to keep you going if you're starting to feel fatigued.

7. Choose multiple performance goals

While not a mental strategy per se, it is advantageous to have more than one goal. By having only one goal (ex: finish-time goal), it is easy to classify the race as either a success or failure. However, having multiple goals during an event enables you to have several points throughout the race to focus on. Such goals are:

- Feel good at mile 10

- Arrive at mile five around 40 minutes
- Stay light on my feet up the hill at mile eight

8. Remember why you're running

Are you running for a charity? Are your kids waiting for you at the finish line? Are you running for health reasons?

Many reasons for running and racing are not directly linked to race-day performance. Remember these reasons and always think big picture!

❖ SUMMARY

- The mental aspect of training has a profound impact on physical performance.
- Mental training is commonly used for:
 - Familiarization with the environment (especially new environments, such as race venues)
 - Motivation
 - Skill development
 - Reduction of negative thoughts and anxiety
 - Establish or reestablish focus
 - Visualize success
- Mental imagery creates the same brain neural patterns as physical activity.
- Confidence can be established and developed through training.
- Like physical training, the quality of imagery correlates directly to its effectiveness.
- External imagery is represented by individuals “seeing” themselves performing an activity.
- Internal imagery is represented by individuals visualizing themselves performing an activity from their own standpoint.
- Mental imagery is most beneficial when done in conjunction with associated physical activity.
- Mental fatigue can negatively influence physical performance.
- A popular way to concentrate during training and racing is to focus on pace.
- Mental burnout exists when an athlete has no desire to train or compete.
 - By tracking a client’s motivation using running logs and adjusting the program accordingly, the chance for burnout can be reduced.
 - Mental burnout is not necessarily a function of the intensity and/or duration of a training program.
- Desensitization is a technique to reduce and/or eliminate phobias through repeated and increased exposure to a phobia.
- Athletes and nonathletes have similar pain threshold levels. However, athletes are typically better at tolerating pain than nonathletes.
- Music can increase the performance of athletes.
- Individuals respond differently to music in regard to sports performance.
- Runners often adopt coping strategies to make it through runs and races.